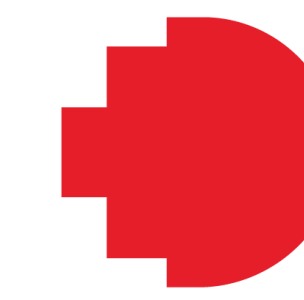


MedPathAgent: Knowledge Graph Augmented Tool-use Medical Agent

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I. Introduction

Large language models (LLMs) offer powerful capabilities for biomedical AI but their tendency to hallucinate [1] makes robust knowledge grounding essential for safe and reliable medical use. Hence it is important to give focus to knowledge grounding when utilizing LLMs for such sensitive domains. There are key grounding techniques:

- **Tool-augmented Reasoning** [2][3] can leverage structured knowledge but has no method of utilising the relations between key entities in text.
- **Knowledge Graph (KG) Augmented Reasoning** [4][5] makes use of the mechanistic and causal relations between entities but has no further reliable context about their accurate meaning and relies on internal model knowledge for this.

To this end our main research question and contributions are as follows:

1. How does augmenting tool-use agents with information from knowledge-graph verified reasoning paths affect their performance in biomedical question-answering tasks? We introduce **MedPathAgent** and show that the knowledge-graph verified paths increase accuracy of tool-use agents.
2. Are knowledge-grounded methods more robust to increase in problem complexity compared to pure LLMs? Here problem complexity will be decided by the average length of top-k reasoning paths to get to the answer node.

II. Method

The method has 2 parts: knowledge graph **path extraction** and **tool-augmented reasoning**. We also compare with key internal reasoning methods: Chain of Thought (CoT), Chain of Verification (CoVe) and Mixture of Experts.

MedPathAgent begins with a biomedical question and its answer options.

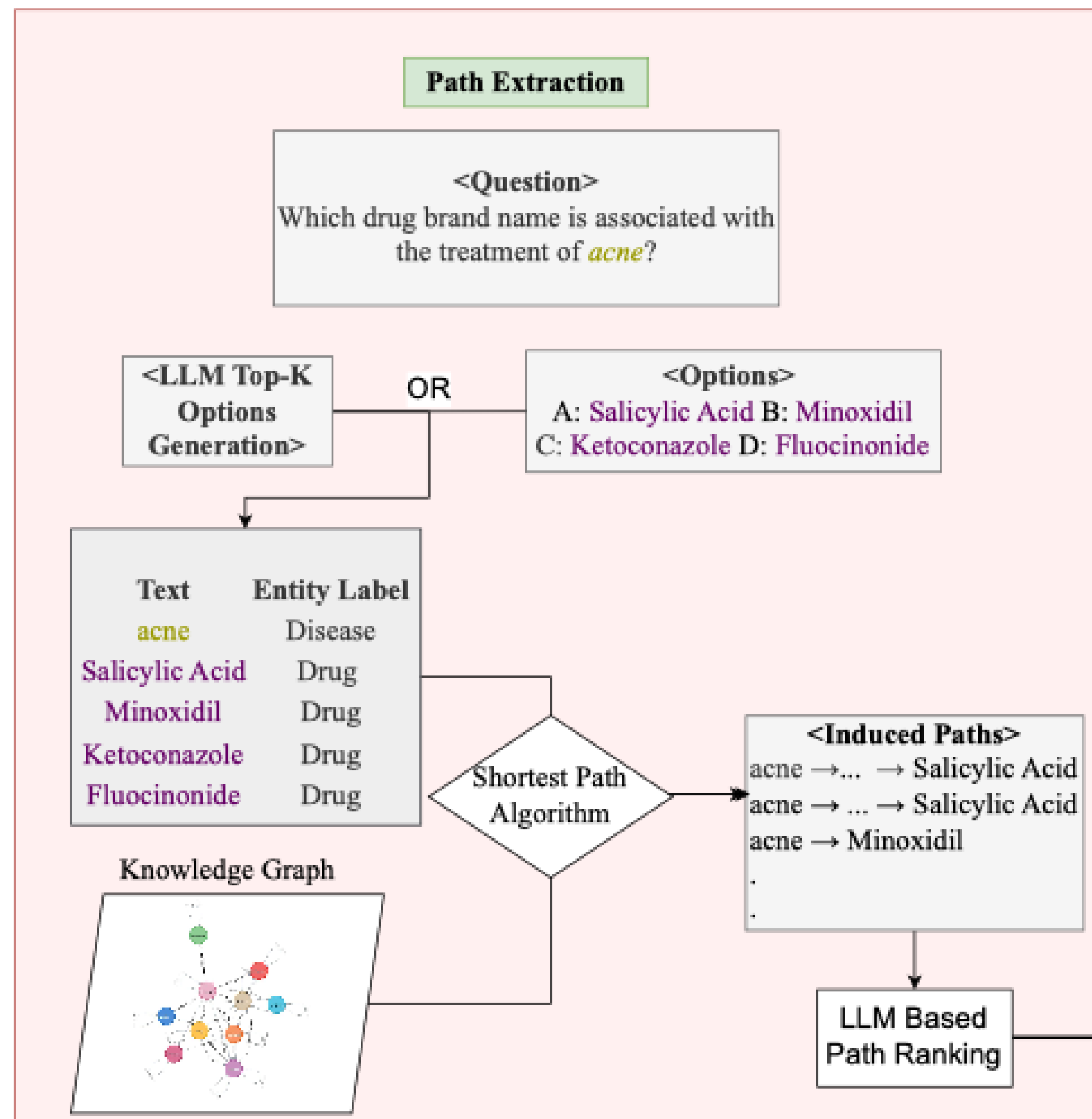
Input Parsing: Identify biomedical entities from both question and answer options using Named Entity Recognition.

Entity Linking: Embed entity mentions using SapBERT [6] and retrieve nearest biomedical concepts from PrimeKG [7] via FAISS K-Nearest Neighbours matching.

Subgraph Induction: Expand linked entities into a local induced subgraph using similarity-based nearest neighbour expansion.

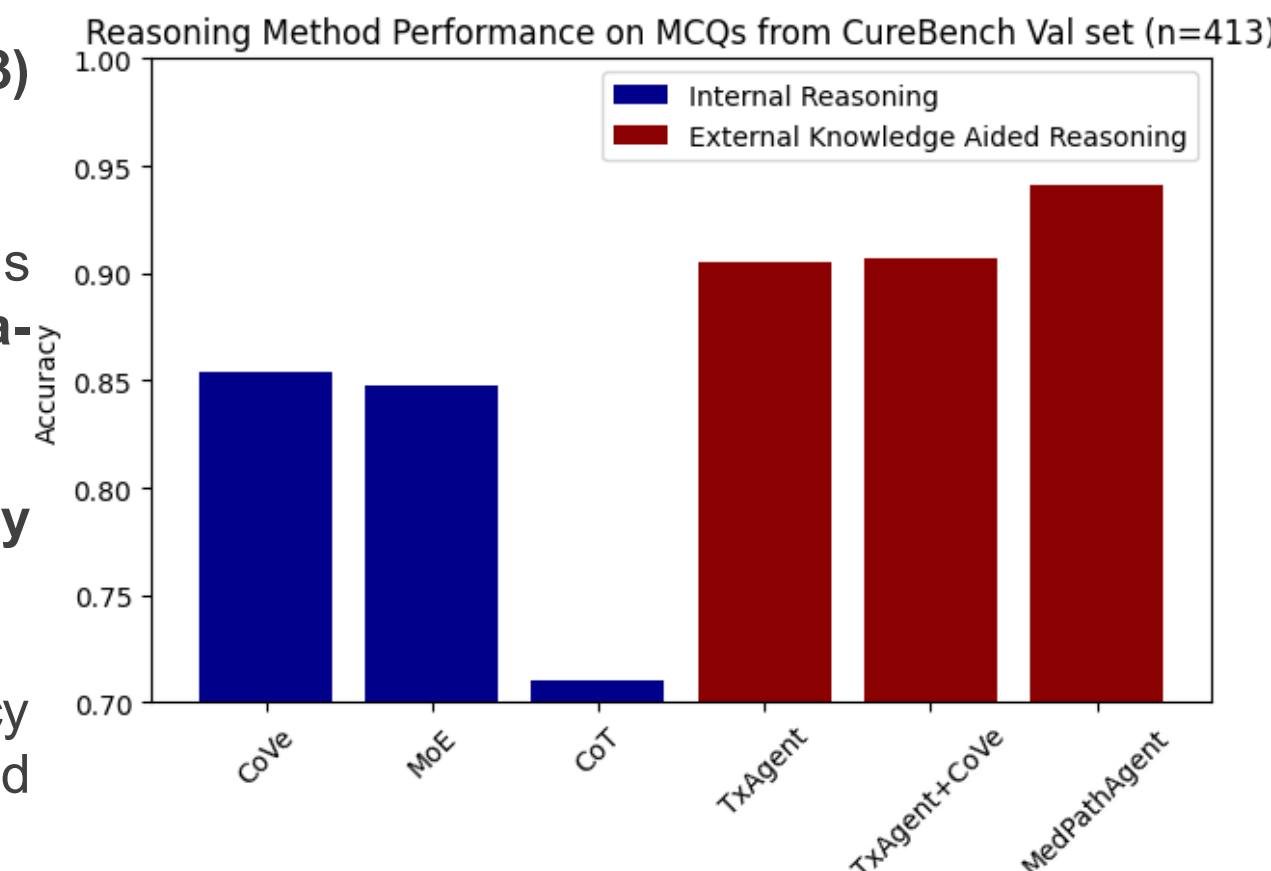
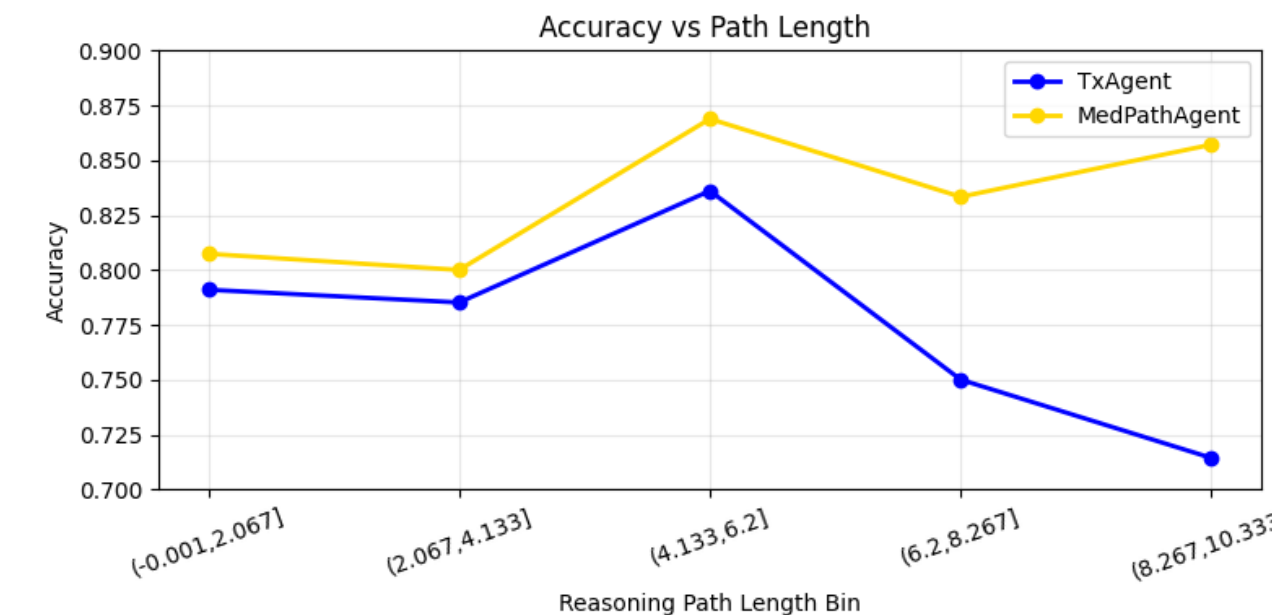
Path Generation: Perform shortest-path discovery within the induced subgraph, extracting both node and edge relations.

Path Ranking: Use an LLM-based ranker to select Top-K clinically relevant reasoning paths aligned with the question context.



III. Experiments

- CureBench [8] **MCQs** from the **validation dataset (n=413)** used for top-1 accuracy evaluation across methods.
- **Llama-3.3-70B** used for all internal reasoning models while knowledge-aided methods use a finetuned **Llama-3.1-8B** combined with **Qwen2-1.5B**.
- External knowledge-aided methods **significantly outperform** internal reasoning methods
- **CoVe** and **MedPathAgent** achieves the highest accuracy for internal reasoning, and external DB-augmented reasoning respectively



- In the validation set 240 samples (out of 459) do not have any paths extracted from the described method.
- TxAgent is **more sensitive to increasing reasoning path complexity**, showing larger dip for longer paths.
- In contrast, MedPathAgent remains **relatively stable**, indicating some ability to exploit deeper relational evidence for improved decision-making.

IV. Next Steps

- Do accuracy gains translate to **lower hallucination metrics** as well?
- How the methods differ in performance based on **different domains** in the dataset?
- Is **ToolRAG** better than singular larger **LLM for both tool retrieval and QA**? This is useful since it is expensive to fit 2 separate LLMs in memory.
- What kind of examples **do not have KG paths present**? Preliminary qualitative analysis shows some queries simple not have the nature which require a "path"; deeper analysis is needed.

In conclusion, tool-augmented reasoning has opened a space for more reliable and trustworthy deployment of LLMs in the drug reasoning domain and combined with knowledge graphs, MedPathAgent provides a technique to ensure robustness across longer KG chains.

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